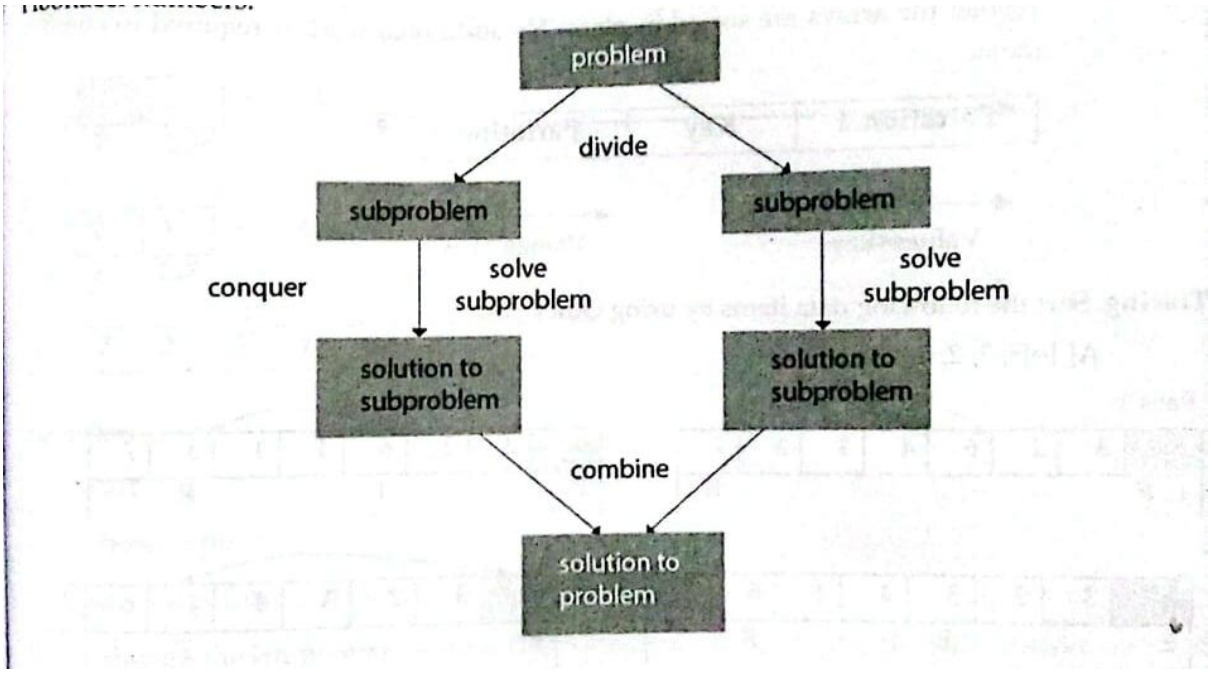
Divide-and-Conquer Algorithm

An important problem-solving technique that uses the concept of recursion is the divide-and-conquer approach. It is an algorithmic design paradigm in which a problem is solved by recursively breaking it into smaller, similar sub-problems, solving each independently, and then combining their solutions to obtain the final result for the original problem.



Using this approach, the main problem is divided into smaller sub-problems, each representing a portion of the original problem. This division continues until the sub-problems become so small that they cannot be divided further (atomic level). These smallest sub-problems are then solved directly. Finally, their solutions are combined to produce the solution to the original problem.

Broadly, the divide-and-conquer approach consists of three main steps:

1. **Divide (Break):**

   The problem is recursively divided into smaller sub-problems until no further division is possible. Each sub-problem represents a part of the original problem.

2. **Conquer (Solve):**

At this stage, each smallest sub-problem is solved individually and directly, as it is simple and cannot be divided any further.

3. **Combine (Merge):**

   The solutions of the sub-problems are recursively combined to form the solution to the original problem. In practice, the conquer and combine steps are closely related and often appear as a single phase.

Comparative Study of Time and Space Complexity of Sorting Algorithms

Sorting Algorithm	**Time Complexity**			**Space Complexity**
	Best Case	**Average Case**	**Worst Case**	
Bubble Sort	O(n)	O(n2)	O(n2)	O(1)
Selection Sort	O(n2)	O(n2)	O(n2)	O(1)
Insertion Sort	O(n)	O(n2)	O(n2)	O(1)
Merge Sort	O(n log(n))	O(n log(n))	O(n log(n))	O(n)
Quick Sort	O(n log(n))	O(n log(n))	O(n2)	O(n)
Heap Sort	O(n log(n))	O(n log(n))	O(n log(n))	O(1)
Shell Sort	O(n log(n))	O(n log(n))	O(n2)	O(1)